**Weekly Report**

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**Project:** UAV-Based Traffic Safety & Video Analytics
**Week:** 14 Mar 2026 – 21 Mar 2026
**Submission Date:** 21 Mar 2026

**Detailed Summary of Work Done During Current Week**

**A. Zone Alignment Verification**

- Continued validation of predefined polygon zones across different segments of the UAV traffic video.
- Verified that zone boundaries remain correctly aligned with the road structure throughout the selected frames.
- Checked consistency of zone overlays for use in later vehicle movement and behavioral analysis.
- Confirmed that the zone definitions are suitable for trajectory-based evaluation.

**B. Semi-Automated Annotation Pipeline Progress**

Work this week focused on further improving the inference-assisted annotation workflow for large-scale UAV video data.

**1. Inference Output Verification**

- Ran object detection inference on selected video segments using the trained YOLO model.
- Verified generated detections, including class labels, confidence values, and bounding box coordinates.
- Checked frame-wise detection outputs for suitability in downstream annotation steps.

**2. CSV-Based Annotation Data Handling**

- Continued using structured CSV files to store frame-wise detection outputs.
- Verified consistency of CSV entries for frame number, bounding box coordinates, class labels, and confidence scores.
- Improved organization of detection data for easier conversion and validation.

### 3. CSV to XML Conversion Testing

- Continued refining the CSV-to-XML conversion process for compatibility with CVAT.
- Tested generated XML files to ensure they follow the expected annotation structure.
- Verified that the converted annotation files can be imported into CVAT for review and correction.

### 4. CVAT-Based Refinement Workflow

- Continued testing the semi-automated annotation pipeline in CVAT.
- Verified that imported bounding boxes can be corrected efficiently instead of manually drawing annotations from scratch.
- Evaluated the practicality of this workflow for handling long video sequences with large numbers of frames.

### C. Vehicle Trajectory Extraction Development

- Continued trajectory extraction using centroid computation from bounding box coordinates.
- Generated trajectory points for selected vehicles over additional frame segments.
- Visualized preliminary vehicle paths to verify movement continuity.
- Observed that maintaining consistent vehicle identity across frames remains a challenge, especially in dense traffic scenarios.
- Continued analyzing how trajectory outputs can be structured for later traffic behavior analysis.

### D. Workflow Structuring and Data Organization

- Continued organizing the overall project pipeline into sequential stages: detection, annotation, refinement, and trajectory extraction.
- Structured intermediate files and outputs for easier validation and future reproducibility.
- Maintained project data in a form suitable for later integration with zone-based behavioral analysis.

## Tasks Planned for Coming Week

### A. Annotation Workflow Finalization

- Further improve CSV to XML conversion reliability.
- Continue validating CVAT import and correction workflow on larger frame segments.
- Refine the semi-automated annotation process for better scalability.

**B. Trajectory Extraction Improvement**

- Extend trajectory generation to longer video segments.
- Improve continuity of vehicle identity across frames.
- Generate more stable visualizations of vehicle movement paths.

**C. Preparation for Behavioral Analysis**

- Begin relating extracted trajectories to predefined zones.
- Prepare structured outputs that can support later lane compliance and violation analysis.
- Continue documenting the complete workflow for project reporting.